H.2. Proof of Task Generalization Failure Threshold

Proof. We establish the exponential decay bound through a probabilistic analysis of reasoning failure modes in the presence of task generalization complexity.

Let Ω denote the sample space of all possible reasoning configurations, and let $C \in \Omega$ represent a specific configuration. We define the following events: A_i as the event that element a_i is novel, i.e., $a_i \notin \mathcal{E}^i_{\text{train}}$; F_j as the event that transformation f_j is novel, i.e., $f_j \notin \mathcal{F}_{\text{train}}$; and Q as the event that the transformation sequence $(f_1, f_2, \ldots, f_k)$ is novel, i.e., $(f_1, f_2, \ldots, f_k) \notin \mathcal{P}_{\text{train}}$.

Here we make the assumption that the reasoning failures induced by novel arguments, functions, and patterns contribute independently to the overall failure probability and hence we model the success probability as a product of component-wise success rates:

$$P(C) = P_0 \prod_{i=1}^{m} \rho_a^{\mathbb{1}[A_i]} \prod_{j=1}^{n} \rho_f^{\mathbb{1}[F_j]} \rho_p^{\mathbb{1}[Q]} \rho_c^{C_T} \tag{30}$$

where $P_0 \in (0, 1]$ represents the baseline success probability when all components are within the training distribution, and $\rho_a, \rho_f, \rho_p, \rho_c \in (0, 1)$ are the degradation factors associated with novel arguments, functions, patterns, and task-specific complexity, respectively.

$$\ln P(C) = \ln P_0 + \sum_{i=1}^{m} \mathbb{1}[A_i] \ln \rho_a + \sum_{j=1}^{n} \mathbb{1}[F_j] \ln \rho_f + \mathbb{1}[Q] \ln \rho_p + C_T \ln \rho_c \tag{31}$$

For notational convenience, we define the positive constants:

$$\begin{aligned} \xi_a &:= -\ln \rho_a > 0, \ \xi_f := -\ln \rho_f > 0, \\ \xi_p &:= -\ln \rho_p > 0, \ \xi_c := -\ln \rho_c > 0 \end{aligned} \tag{32}$$

hence we have:

$$\begin{aligned} \ln P(C) = \ln P_0 - \xi_a \sum_{i=1}^{m} \mathbb{1}[A_i] - \xi_f \sum_{j=1}^{n} \mathbb{1}[F_j] \\ - \xi_p \mathbb{1}[Q] - \xi_c C_T \end{aligned} \tag{33}$$

Lemma: Relationship to TGC. The expression in equation above can be bounded in terms of TGC(C) as follows:

$$\ln P(C) \leq \ln P_0 - \delta \cdot \text{TGC}(C) \tag{34}$$

where $\delta = \min(\frac{\xi_a}{\alpha}, \frac{\xi_f}{\beta}, \frac{\xi_p}{\gamma}, \xi_c) > 0$.

Proof of Lemma: From the definition of TGC(C) in Eq. (14), we have:

$$\text{TGC}(C) = \alpha \sum_{i=1}^{m} \mathbb{1}[A_i] + \beta \sum_{j=1}^{n} \mathbb{1}[F_j] + \gamma \mathbb{1}[Q] + C_T \tag{35}$$

By the definition of δ, each term in Eq. (33) satisfies:

$$\xi_a \sum_{i=1}^{m} \mathbb{1}[A_i] \geq \delta\alpha \sum_{i=1}^{m} \mathbb{1}[A_i] \tag{36}$$

$$\xi_f \sum_{j=1}^{n} \mathbb{1}[F_j] \geq \delta\beta \sum_{j=1}^{n} \mathbb{1}[F_j] \tag{37}$$

$$\xi_p \mathbb{1}[Q] \geq \delta\gamma \mathbb{1}[Q] \tag{38}$$

$$\xi_c C_T \geq \delta C_T \tag{39}$$

Summing these inequalities establishes Eq. (34).

We now define the threshold $\tau := \frac{\ln P_0}{\delta}$. From Eq. (34), when $\mathrm{TGC}(C) > \tau$, we have:

$$\begin{aligned} \ln P(C) &\leq \ln P_0 - \delta \cdot \mathrm{TGC}(C) \\ &= \delta(\tau - \mathrm{TGC}(C)) \\ &= -\delta(\mathrm{TGC}(C) - \tau) \end{aligned} \tag{40}$$

Exponentiating both sides yields the desired bound: $P(C) \leq e^{-\delta(\mathrm{TGC}(C)-\tau)}$ □

H.3. Proof of Length Extrapolation Bound

Proof. Consider a transformer model f_θ processing sequences of length L. The model implicitly learns position-dependent representations through positional encodings $\mathrm{PE}(i) \in \mathbb{R}^d$ for position $i \in \{1, \ldots, L\}$ and attention patterns $A_{ij} = \mathrm{softmax}\left(\frac{Q_i K_j^T}{\sqrt{d}}\right)$.

During training on fixed length L_{train}, the model learns a specific distribution:

$$p_{\text{train}}(\mathbf{h}) = p(\mathbf{h} \mid L = L_{\text{train}}) \tag{41}$$

where $\mathbf{h} = \{h_1, \ldots, h_L\}$ represents hidden states.

For sequences of length $L \neq L_{\text{train}}$, we encounter distribution shift in two forms: (1) positional encoding mismatch, where the model has never seen positions $i > L_{\text{train}}$ if $L > L_{\text{train}}$, and (2) attention pattern disruption, where the learned attention patterns are calibrated for length L_{train}.

The KL divergence between training and test distributions can be bounded:

$$D_{KL}(p_{\text{test}} \| p_{\text{train}}) \propto |L - L_{\text{train}}|^2 \tag{42}$$

This quadratic relationship arises from linear accumulation of positional encoding errors and quadratic growth in attention pattern misalignment due to pairwise interactions.

Let $\mathcal{E}(L)$ be the prediction error at length L. We decompose it as:

$$\mathcal{E}(L) = \mathcal{E}_{\text{inherent}}(L) + \mathcal{E}_{\text{shift}}(L) \tag{43}$$

where $\mathcal{E}_{\text{inherent}}(L) = \mathcal{E}_0$ is the inherent model error (constant) and $\mathcal{E}_{\text{shift}}(L)$ is the error due to distribution shift.

The distribution shift error follows from the Central Limit Theorem. As the error accumulates over sequence positions, the total shift error converges to:

$$\mathcal{E}_{\text{shift}}(L) = (1 - \mathcal{E}_0) \cdot \left(1 - \exp\left(-\frac{(L - L_{\text{train}})^2}{2\sigma^2}\right)\right) \tag{44}$$

This form ensures that $\mathcal{E}_{\text{shift}}(L_{\text{train}}) = 0$ (no shift at training length) and $\lim_{|L-L_{\text{train}}|\to\infty} \mathcal{E}_{\text{shift}}(L) = 1 - \mathcal{E}_0$ (maximum error bounded by 1).

The width parameter σ depends on:

$$\sigma = \sigma_0 \cdot \sqrt{\frac{d}{L_{\text{train}}}} \tag{45}$$

where σ_0 is a model-specific constant, d is the model dimension, and the $\sqrt{d/L_{\text{train}}}$ factor captures the concentration of measure in high dimensions.

Therefore, the total error follows:

$$\mathcal{E}(L) = \mathcal{E}_0 + (1 - \mathcal{E}_0) \cdot \left(1 - \exp\left(-\frac{(L - L_{\text{train}})^2}{2\sigma^2}\right)\right) \tag{46}$$

This Gaussian form naturally emerges from the accumulation of position-dependent errors and matches the experimental observation of near-zero error at $L = L_{\text{train}}$ with symmetric increase in both directions. □

I. Discussion and Implication

Our investigation, conducted through the controlled environment of DataAlchemy, reveals that the apparent reasoning prowess of Chain-of-Thought (CoT) is largely a brittle mirage. The findings across task, length, and format generalization experiments converge on a conclusion: CoT is not a mechanism for genuine logical inference but rather a sophisticated form of structured pattern matching, fundamentally bounded by the data distribution seen during training. When pushed even slightly beyond this distribution, its performance degrades significantly, exposing the superficial nature of the "reasoning" it produces.

While our experiments utilized models trained from scratch in a controlled environment, the principles uncovered are extensible to large-scale pre-trained models. We summarize the implications for practitioners as follows.

Guard against over-reliance and false confidence. CoT should not be treated as a "plug-and-play" module for robust reasoning, especially in high-stakes domains like medicine, finance, or legal analysis. The ability of LLMs to produce "fluent nonsense"—plausible but logically flawed reasoning chains—can be more deceptive and damaging than an outright incorrect answer, as it projects a false aura of dependability. Sufficient auditing from domain experts is indispensable.

Prioritize OOD testing. Standard validation practices, where the test set closely mirrors the training set, are insufficient to gauge the true robustness of a CoT-enabled system. Practitioners must implement rigorous **adversarial and OOD testing** that systematically probes for vulnerabilities across task, length, and format variations.

Recognize fine-Tuning as a patch, not a Panacea. Our results show that Supervised Fine-Tuning (SFT) can quickly "patch" a model's performance on a new, specific data distribution. However, this should not be mistaken for achieving true generalization. It simply expands the model's "in-distribution" bubble slightly. Relying on SFT to fix every OOD failure is an unsustainable and reactive strategy that fails to address the core issue: the model's lack of abstract reasoning capability.

J. Use of Generative AI

To enhance clarity and readability, we utilized the GPT-5.2 model exclusively as a language polishing tool. Its role was confined to proofreading, grammatical correction, and stylistic refinement—functions